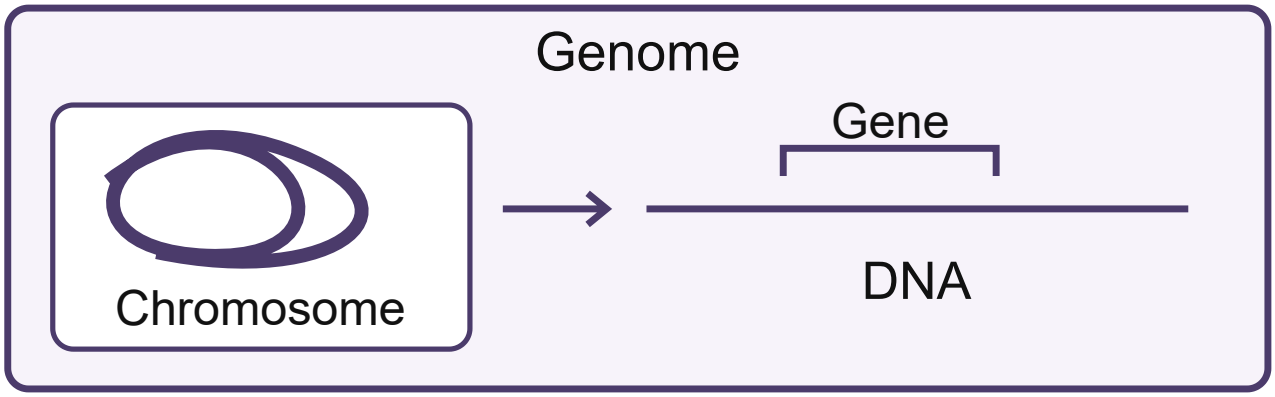
Reference A chromosome model



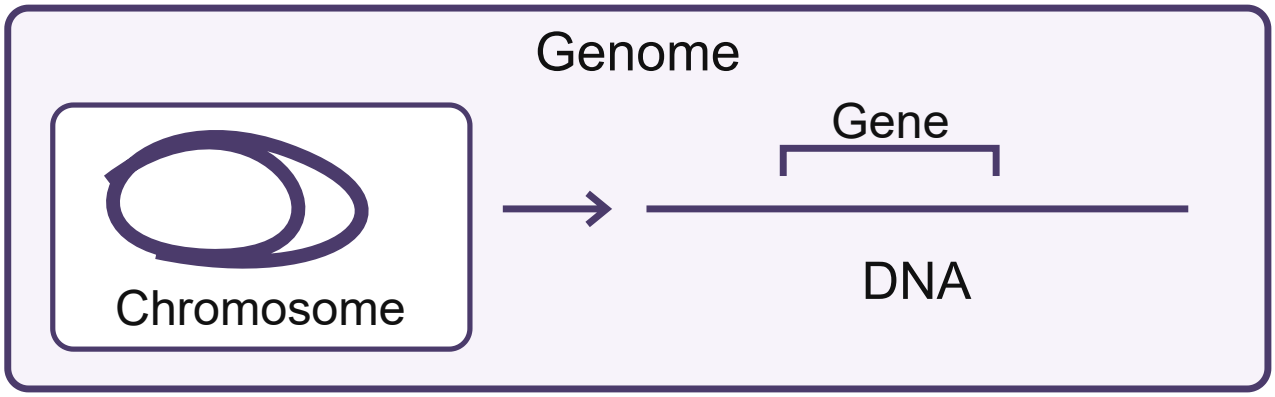
The genome frame surrounds a chromosome model. A cord represents DNA and a bracket marks a gene.

Reference only. This simplified drawing does not show scale.

Part of model	What it represents
Outside frame	Genome. All the DNA of the organism.
Coiled package	Chromosome. A long DNA molecule packaged with proteins.
Cord	DNA. The molecule that stores genetic information.
Bracket	Gene. One region of DNA.

Made by Matt

Reference B explain the links



The frame stands for the genome, including DNA from all chromosomes.

Reference only.

Link	Explanation
Gene and DNA	A gene is a section of DNA.
DNA and chromosome	One chromosome contains a long DNA molecule packaged with proteins.
Chromosome and genome	One chromosome carries part of the full DNA set.

The picture shows only one chromosome in detail. The genome includes all the DNA of the organism.

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Arrival and exit Aspire

Use the picture on Reference A for question 1 and the model table for questions 2–4.

1. Which label points to the short bracketed region?

2. Why can one chromosome contain many genes?

3. Why does the outside frame represent a larger set?

4. Name one thing the string model leaves out.

Exit ticket

1. Repair the label “gene” placed on the whole DNA cord.

2. Name one way to make the model clearer.

Pupil voice

What helped you most today, or what would you change next time?

Point, speak, sign, draw, write or choose. You may pass.

verbal · written · drawn · signed · symbol / choice-selected · demonstrated / re-attempt · adult-scribed. You may pass.

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You do Aspire

Use your book or this sheet. Use Reference A and B. One response per task.

1. Compare the role of a chromosome with the role of a protein-coding gene.
Staff I ☐ S ☐ C ☐

---

2. Correct both errors in a drawing that puts DNA inside a gene and all genes on one chromosome.
Staff I ☐ S ☐ C ☐

---

3. Defend one edit to make the string model more accurate.
Staff I ☐ S ☐ C ☐

Feedback and response · Aspire

Feedback codes

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

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